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In [ ]:
```
from google.colab import files
uploaded = files.upload()
```

In [ ]:
```
import pandas as pd
df=pd.read_csv('Iris.csv')
df
```

Out[ ]:

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Speci
0	1	5.1	3.5	1.4	0.2	Ir setc
1	2	4.9	3.0	1.4	0.2	Ir setc
2	3	4.7	3.2	1.3	0.2	Ir setc
3	4	4.6	3.1	1.5	0.2	Ir setc
4	5	5.0	3.6	1.4	0.2	Ir setc
...	...	...	...	...	...	
145	146	6.7	3.0	5.2	2.3	Ir vir gin
146	147	6.3	2.5	5.0	1.9	Ir virgin
147	148	6.5	3.0	5.2	2.0	Ir vir gin
148	149	6.2	3.4	5.4	2.3	Ir vir gin
149	150	5.9	3.0	5.1	1.8	Ir virgin

150 rows × 6 columns

Mean

In [ ]:
```
df['SepalLengthCm'].min()
```

Out[ ]: 4.3

Median

In [ ]:
```
df['PetalWidthCm'].median()
```

Out[ ]: 1.3

Mode

In [ ]:
```
df['SepalLengthCm'].mode()
```

Out[ ]:

	SepalLengthCm
0	5.0

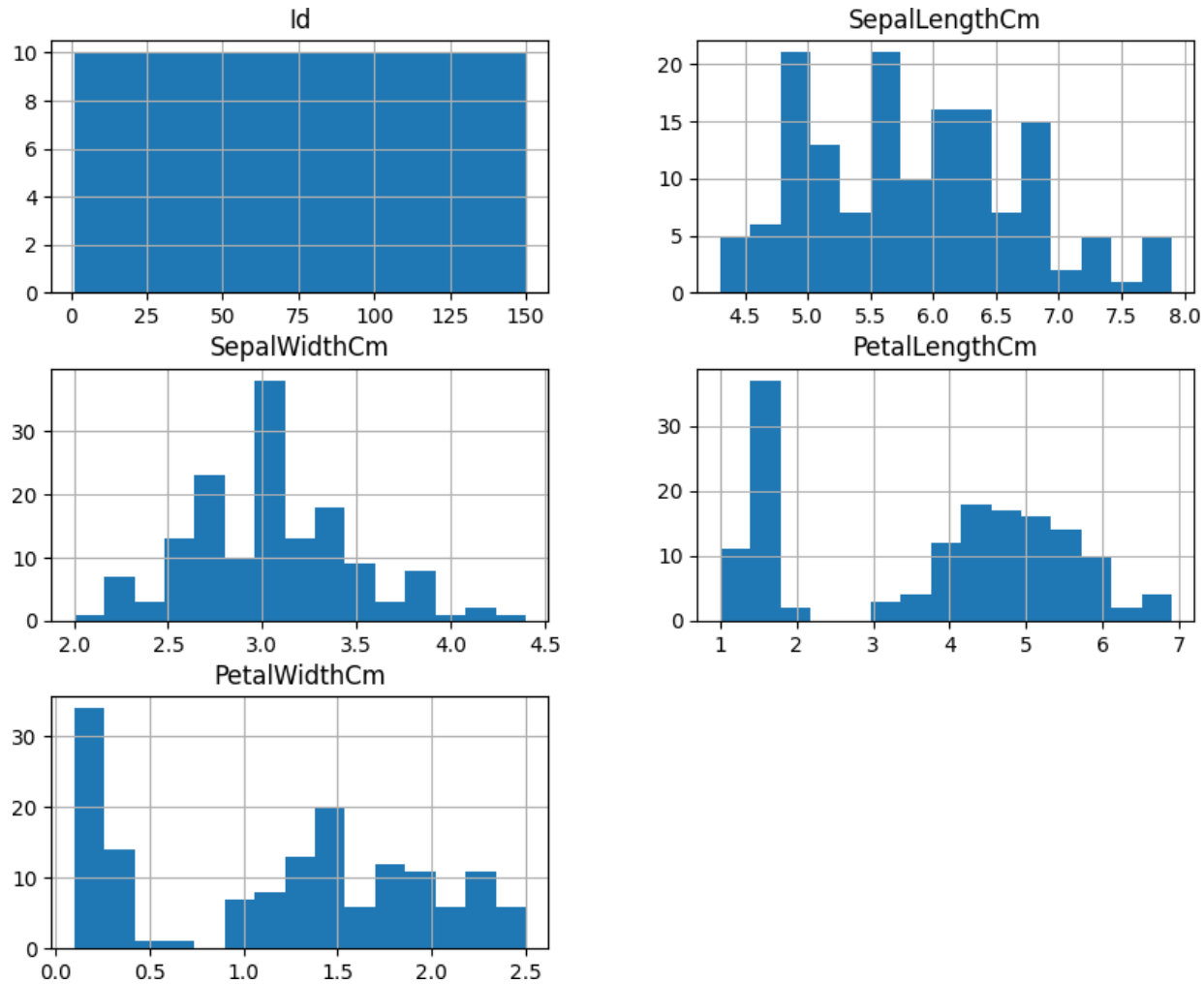
dtype: float64

Feature Distribution using Histogram

In [ ]:
```
df.iloc[:, :-1].hist(figsize=(10,8), bins=15)
plt.suptitle("Histogram of Iris Dataset Features")
plt.show()
```

```
---------------------------------------------------------------------------
NameError                                 Traceback (most recent call last)
/tmp/ipython-input-1914902051.py in <cell line: 0>()
      1 df.iloc[:, :-1].hist(figsize=(10,8), bins=15)
----> 2 plt.suptitle("Histogram of Iris Dataset Features")
      3 plt.show()

NameError: name 'plt' is not defined
```



Data Cleaning

In [ ]:
```
df.isnull().sum()
df.drop_duplicates(inplace=True)
print("Shape after removing duplicates:", df.shape)
```

```
Shape after removing duplicates: (150, 6)
```

Data Integration

In [ ]:
```
X = df[['SepalLengthCm', 'SepalWidthCm',
        'PetalLengthCm', 'PetalWidthCm']]

y = df['Species']
```

Data Transformation

In [ ]:
```
from sklearn.preprocessing import LabelEncoder
le = LabelEncoder()
df['Species_encoded'] = le.fit_transform(df['Species'])
```

Data Model Building (Classification) Create Target Class

In [ ]:
```
features = ['SepalLengthCm',
            'SepalWidthCm',
            'PetalLengthCm',
            'PetalWidthCm']
```

In [ ]:
```
X = df[features]
y = df['Species_encoded']
```

Prepare Training Data

In [ ]:
```
['Id', 'SepalLengthCm', 'SepalWidthCm',
 'PetalLengthCm', 'PetalWidthCm', 'Species']
```

Out[ ]:
```
['Id',
 'SepalLengthCm',
 'SepalWidthCm',
 'PetalLengthCm',
 'PetalWidthCm',
 'Species']
```

Train-Test Split

In [ ]:
```
from sklearn.model_selection import train_test_split

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
```

Apply Classification Model

In [ ]:
```
from sklearn.linear_model import LogisticRegression

model = LogisticRegression()
model.fit(X_train, y_train)
y_pred = model.predict(X_test)
```

Model Evaluation

In [ ]:
```
from sklearn.metrics import accuracy_score, classification_report

print("Accuracy:", accuracy_score(y_test, y_pred))
print(classification_report(y_test, y_pred))
```

```
Accuracy: 1.0
              precision    recall  f1-score   support

           0       1.00      1.00      1.00        10
           1       1.00      1.00      1.00         9
           2       1.00      1.00      1.00        11

    accuracy                           1.00        30
   macro avg       1.00      1.00      1.00        30
weighted avg       1.00      1.00      1.00        30
```